COMPUTER SCIENCE 475
COMPUTATION THEORY

UNIT #1 STUDY GUIDE

GIVEN: $a^+|(a^+b|b)(a|b)^*$ AND GIVEN: $(a^+|\ ab^*\ |\ (cc)^+)^*$

PROBLEM #1

DRAW AN APPROPRIATE NFA.

PROBLEM #2

DERIVE THE BTM CLOSURE FOR THE LAMBDA TRANSITIONS.

PROBLEM #3

DERIVE THE STATE TRANSITION TABLE.

PROBLEM #4

MINIMIZE.

PROBLEM #5

WRITE THE PRODUCTION RULES.

PROBLEM #6

WRITE THE PSEUDO-CODE.

PROBLEM #7

USE YOUR PSEUDO-CODE TO TRACE aba;

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UNIT #1 FOUR POINT PROBLEM SET

GIVEN: $(a^* b^+) \mid (a^+ b^*) \mid (a^* \mid b^*)$

PROBLEM #1

DRAW AN APPROPRIATE NFA.

PROBLEM #2

DERIVE THE BTM CLOSURE FOR THE LAMBDA TRANSITIONS.

PROBLEM #3

DERIVE THE STATE TRANSITION TABLE.

PROBLEM #4

MINIMIZE. NOTE: THERE MAY BE NONE.

PROBLEM #5

WRITE THE PRODUCTION RULES.

PROBLEM #6

WRITE THE PSEUDO-CODE. NOTE: A to B on a becomes A→aB

PROBLEM #7

USE YOUR PSEUDO-CODE TO TRACE ab#